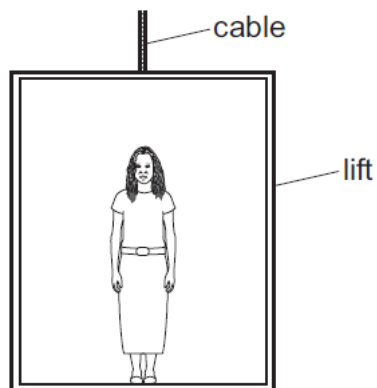
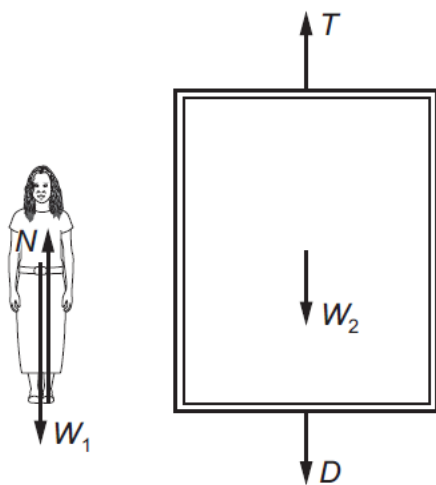


- 5 The diagram shows a woman standing in a lift.



The forces acting on the woman and the forces acting on the lift are shown.



N is the force from the lift floor on the woman.

W_1 is the weight of the woman.

T is the tension in the lift cable.

W_2 is the weight of the lift.

D is the force from the woman on the lift floor.

Which statement is correct?

- A N and W_1 are always equal and opposite.
- B $(W_1 + W_2)$ is always equal to T .
- C If $N = W_1$, the lift must be at rest.
- D If $T = (D + W_2)$, the lift must have a constant velocity.

6

Ans: **D**

6 A cylindrical drum has radius r and weight W . The drum is to be rolled over onto a step of height